

Jacob Mattie

Curriculum Vitae

Vancouver | 778-710-7554 | j_mattie@live.ca
<https://github.com/qimbet> | www.qimbet.com

Summary

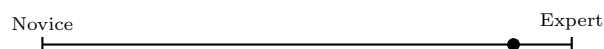
Mathematician with vocational expertise in mechanical engineering and software development. Automation-focused skillset. Specialized in data pipelines & databases.

My academic goals are to pursue a Master's Degree and a PhD in Molecular Physics.

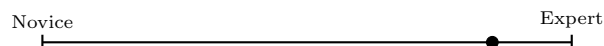
Skills

Programming Languages:

Python



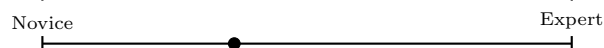
SQL



Bash



C++

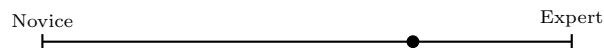


Fabrication:

Microcontrollers, Electronic Design, & Assembly



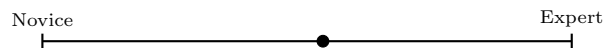
Plumbing, Pumps, & Valves



Sensors, IoT, Device Networking



120+ VAC Electronics

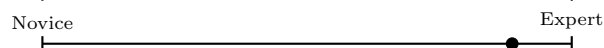


Data Engineering:

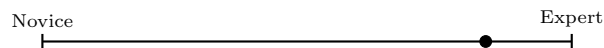
Data Structures



Data Analysis

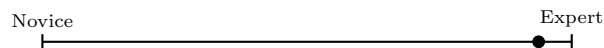


Algorithm Design

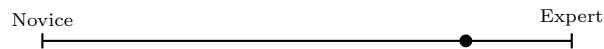


Software & Frameworks :

Microsoft Office (Excel, Word, PowerPoint)



Linux



SolidWorks, FreeCAD, AutoCAD

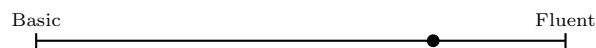


Django



Languages:

French



Work Experience

Investment Analyst

Aligned Capital Partners / River Wealth

Vancouver, BC
11/2024 - Ongoing

- Built a Django web application from scratch – from Linux server configuration to **database schema design** and GUI development.
- Created a suite of fully-automated ETL (Extract, Transform, Load) **data pipelines** paired with a task scheduler to grow a time-series archive for over 23,000 global securities, client data, and third-party values, enabling quantitatively-supported investment decisions.
- Automated recurring tasks in Python, reducing office-wide labour load by 20% and mitigating possibility of human error.

General Contractor

Independent

Calgary, AB
6/2024 – 8/2025

Retail Clerk

Fair's Fair Books

Calgary, AB
1/2024 - 8/2024

Test Technologist

IBC Technologies

Burnaby, BC
6/2022 – 9/2023

- Led compliance testing on pre-release boiler units.
- Automated hardware testing with IoT microcontrollers and PLC devices, creating **cloud-connected data pipelines** to asynchronously run validation tests.
- **Automated data analysis** of million-point datasets using MATLAB, improving efficiency in test validation by over 600% .
- Designed and deployed laser-based obstruction sensors and a responsive control system to an **assembly line**, boosting station turnover speed by 25% while simultaneously enhancing operator safety.

Opinions Editor, Contributor

The Peak Publications Society

Burnaby, BC
10/2020 – 6/2022

- Managed the Opinions Section of a weekly newspaper, coordinating 5 rotating student contributors and staff writers to each publish 500 words in face of production deadlines.
- **Mentored and coached writers** of varying experience levels with thematic guidance and structural training to develop their voice and orchestrate their work with the greater section.

Skilled Labour

HD Concrete

Surrey, BC
2/2021 – 1/2022

IT Technician (Internship)

BC Cancer Research Centre

Vancouver, BC
09/2018 – 01/2019

- Diagnosed and **resolved technical issues** on computers and digital equipment in a large-scale research facility.
- Configured network switches, user accounts, and domain permissions to support **cybersecurity** and preserve data isolation across user groups.
- Wrote internal **documentation** standardizing common tasks and foundational knowledge, facilitating training of future students.

Expeditor Lead, Busser

Steamworks Brewpub

Vancouver, BC
09/2017 – 09/2018

General Labour

Vision Built Homes

Vancouver, BC
4/2016 – 9/2016

Bartender

The Charles Bradlaugh

Northamptonshire, UK
4/2015 – 8/2015

Line Cook

White Spot Restaurants

Burnaby, BC
4/2014 – 4/2015

Retail Clerk <i>JB Hi-Fi</i>	Sydney, AUS 10/2013 – 3/2014
General Labour <i>Hunter Roofing</i>	Surrey, BC 4/2013 – 8/2013
Clerk <i>Price Smart Foods</i>	Burnaby, BC 5/2011 – 8/2012

Projects

Hydroponics 04/2025 – 08/2025; Ongoing
Independent Project

- Designed and assembled a modular hydroponics system able to grow 36 spinach plants from seed to harvest with no manual input.
- Engineered esp32-based **circuitry** with relays, sensors, and voltage converters to control grow lights, watering pumps, and fertilizer dispersion systems.
- Planned integration of a robotic pollinator arm using computer vision techniques to enable growth of a broader range of fruit & berry crops.

ROV – Underwater Surveying 02/2025 – Frozen (06/2025)
Collaborative Project

- Researched market potential for underwater surveying vehicles by talking to professionals in marine industries and regulatory bodies, **qualifying design requirements** and constraints.
- Designed a production-ready, low-cost, modular carrier and signal relay vehicle for the ROV, enabling radio-frequency control for an underwater vehicle.
- Combined off-the-shelf components and custom FreeCAD-designed parts in an **assembly model** for the carrier boat to validate the project's feasibility.
- Developed an **electrical management** and distribution system with capacity appropriate to typical mission length and intensity, minimizing battery cost without sacrificing functionality.

Compressed Air Energy Storage 3/2024 – 8/2024
Startup Co-Founder

- Researched the advent of wind & solar power on global energy grids, and the subsequent opportunities in short-term electrical storage.
- Planned and pitched compressed air as an accessible and achievable energy storage method to investors, power plant owners, and grid administrators.

Brewing 05/2018 - 09/2023
Hobby

- Produced over 1,000 bottles of beer over 22 batches, using self-developed recipes.
- Developed a system of heat exchangers, pumps, airlocks, and valves to monitor and control the temperature, sterility, and filtration of beer throughout the stages of brewing.

Map the System 01/2021 – 03/2024
Research Competition

- In a team of 4, published a 15-page research report and multimedia presentation on the barriers to access healthcare throughout BC.
- Sourced data through interviews with pharmacists, doctors, and healthcare dependents from different backgrounds to understand the problem landscape.

Greenhouse Gas Emissions Audit
Comissioned by: Gabriola Island Futures

08/2020 – 04/2021

- Quantified Greenhouse Gas Emissions for the island community of Gabriola by surveying retail sources, productive sites, and utilities providers.
 - Delivered an emissions footprint calculator, summing gaseous emissions across the supply chain as CO2-equivalent values.
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Education

Simon Fraser University
B.A.Sc Mathematics

01/2019 - 05/2022

Key Coursework:

- Real Analysis
- Algebra II: Rings and Fields
- Continuous Optimization
- Applied Discrete Mathematics

Simon Fraser University
B.A.Sc Engineering - Not Completed

09/2017 - 12/2018

Key Coursework:

- Process of Professional Writing
- Graphical Communications

Langara College
Associate Of Science Certificate - Not Completed

01/2015 - 05/2017

Key Coursework:

- Calculus III
 - Engineering Mechanics
 - Electromagnetism & Optics
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Certifications

Nuclear Energy Worker
Canadian Nuclear Safety Commission

09/2025 - 09/2030

Ocean AI and Technology Foundations
Canada's Ocean Supercluster

Attained: 06/2025

Occupational First Aid - Level 1
Metro Safety Training

10/2021 - 10/2024

Fall Protection
Metro Safety Training

06/2019 - 06/2022

Occupational First Aid - Level 1
Metro Safety Training

10/2014 - 10/2017

Class 5 Driver's License
ICBC

03/2012 - 04/2025

Volunteering & Leadership

TRIUMF: ASPIRE Lab

Volunteer: Automation Technologist

Vancouver, British Columbia
05/2025 - Ongoing

- Contributed to the development of 3 papers:
 - Repeatable EPICS Installer; Project In Progress
 - Control Systems; Paper In Preparation
 - Astrochemistry; Paper In Preparation
- Developed a Bash-based installer for EPICS, an industry-standard **lab control framework** for physics laboratories, mitigating a significant entry labour barrier for new students and staff.
- Programmed a driver for a stepper motor, enabling its integration into the laboratory control system, **improving experimental repeatability** and reducing hands-on labour during beamtime.

Beaty Biodiversity Museum: Vertebrate Lab

Volunteer: Laboratory Assistant

Vancouver, British Columbia
09/2025 - 11/2025

- Performed necropsies on animal carcasses, gathering tissue samples and recording specimen data.
- Collaborated with entomology and botany departments to identify animal stomach contents, improving **data quality** on the species' behaviours.
- Acted as caretaker to dermestid beetle colonies while monitoring their cleaning of sample skeletons.

Epic Fencing Club

Facilities Improvement, Athlete

Calgary, Alberta
4/2025 – 8/2025

- Soldered, crimped, and assembled 40 wiring harnesses for electric reels and pistes.
- Installed **infrastructure** for ceiling-mounted cable reels, reducing wire clutter in a high-traffic space.
- Validated and repaired weapons, body cords, and scorebox equipment during tournaments.

IBC Technologies

Employee Committee – Engineering Representative

Burnaby, British Columbia
03/2023 – 09/2023

- Boosted workplace morale by setting up a community garden near the parking lot, and through a company-wide plant giveaway.
- Organized and facilitated monthly company lunches, feeding over 200 employees.

The Peak Publications Society

Board of Directors – Collective Representative

Burnaby, British Columbia
06/2021 – 12/2021

- Collaborated with financial advisors to manage a \$1.5 million investment account, finding a balance between financial and ethical values.
- Researched, presented, and voted on **governance** issues, including operating budgets and the long-term goals of the newspaper.

SFU Fencing Team

Team Administrator, Athlete

Burnaby, British Columbia
09/2018 – 12/2021

- Raised \$500 for a nascent student athletics team by advertising and running a fencing tournament, enabling the purchase of an electric scoring machine.

SMA Negeri 5 Palangka Raya

English Teacher

Central Kalimantan, Indonesia
1/2013 - 3/2013

- Taught English to high-school-level students who had little prior English-language exposure.